Literature Survey

2.1 List of Research Papers

1. Investigation on Smart Campus Management Platform Based on Digital Twin (Elsevier, 2023)
2. Developing Campus Digital Twin Using Interactive Visual Analytics Approach (Springer, 2024)
3. A Smart Campus' Digital Twin for Sustainable Comfort Monitoring (MDPI Sustainability, 2020)
4. Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study (MDPI Smart Cities, 2024)
5. BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP (Elsevier, 2025)
6. The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization (Springer, 2026)
7. Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence (Elsevier, 2026)
8. Digital Twins: Cornerstone to Circular Economy and Sustainability Goals (Springer, 2025)
9. Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience (Springer, 2026)
10. A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness (MDPI Sustainability, 2026)
11. Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment (MDPI Sustainability, 2026)
12. From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy (Springer, 2023)
13. AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems (MDPI Buildings, 2025)
14. Digital Twin-Enabled Smart Facility Management: A Bibliometric Review (Springer, 2024)
15. A Review of Building Digital Twins to Improve Energy Efficiency (Springer – Energy Informatics, 2024)

2.2 Literature Survey Table

Paper	Method	Dataset	Advantages	Limitations	Research Gap
Investigation on Smart Campus Management Platform Based on Digital Twin	Digital Twin integrated with IoT, BIM, and cloud platform for campus management.	Real-time campus IoT sensor data.	Real-time monitoring, improved campus management, predictive maintenance.	Limited scalability and AI integration.	Incorporate AI-driven prediction, cloud-native architecture, and advanced sustainability analytics.
Developing Campus Digital Twin Using Interactive Visual Analytics Approach	Interactive visual analytics integrated with BIM, GIS, and Digital Twin.	Campus GIS, BIM, and sensor data.	Enhanced visualization and decision-making for campus management.	Dependent on accurate data and lacks predictive analytics.	Integrate AI-based prediction and automated cloud analytics.
A Smart Campus' Digital Twin for Sustainable Comfort Monitoring	IoT-enabled Digital Twin for indoor environmental comfort monitoring.	Temperature, humidity, CO_2, occupancy, and environmental sensor data.	Improves occupant comfort, energy efficiency, and indoor monitoring.	Limited to indoor environments and comfort analysis.	Extend to campus-wide sustainability monitoring with cloud analytics.
Multifunctional Models in Digital and Physical Twinning of the Built Environment—A University Campus Case Study	Combines Digital Twin, BIM, and simulation models for campus management.	University campus BIM and operational datasets.	Comprehensive facility representation and improved maintenance planning.	High computational complexity and deployment cost.	Develop scalable cloud-based Digital Twin with AI-assisted decision support.

BIM-Based Digital Twin Development for University Campus Management: Case Study ETSICCP	BIM integrated with Digital Twin for infrastructure management.	BIM models and campus facility management data.	Better visualization, infrastructure management, and maintenance.	Manual synchronization between BIM and real-world updates.	Automate synchronization using IoT sensors and cloud computing.
The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization	3D Digital Twin modelling for campus planning and optimization.	GIS and CAD datasets.	Improved planning, visualization, and infrastructure design.	Focuses mainly on planning rather than operational monitoring.	Integrate real-time IoT data and sustainability analytics.
Digital Twins for Smart Campus Networks: An End-to-End Framework for Multi-Domain Data Intelligence	Cloud-based Digital Twin integrating networking, IoT, and campus systems.	IoT devices, network traffic, and operational data.	Cross-domain analytics, scalability, and efficient resource utilization.	Complex implementation and interoperability issues.	Improve security, AI automation, and sustainability prediction.
Digital Twins: Cornerstone to Circular Economy and Sustainability Goals	Review of Digital Twin applications supporting sustainability and circular economy.	Multiple published case studies.	Comprehensive overview of Digital Twin technologies.	No practical implementation or validation.	Develop cloud-based Digital Twin applications for educational campuses.

Linking Digital Twin Paradigm for Urban Heat Monitoring and Policy Integration to Building Smart City Climate Resilience	Digital Twin integrated with urban climate monitoring.	Urban environmental and weather datasets.	Supports climate resilience and sustainable planning.	Primarily focuses on smart cities rather than campuses.	Adapt the framework for campus sustainability and energy optimization.
A BIM-Integrated Eco-Digital Framework for Predictive Maintenance and Sustainability Assessment in Educational Buildings Towards Digital Twin Readiness	BIM integrated with Digital Twin for predictive maintenance and sustainability assessment.	Building maintenance and energy consumption datasets.	Predictive maintenance, sustainability assessment, and lifecycle management.	Limited practical deployment in large campuses.	Develop cloud-enabled Digital Twin with AI-based predictive analytics.
Sustainable Energy Transitions in Smart Campuses: An AI-Driven Framework Integrating Microgrid Optimization, Disaster Resilience, and Educational Empowerment	AI-based microgrid optimization for sustainable campus energy management.	Campus energy consumption and renewable energy datasets.	Energy optimization, renewable integration, and disaster resilience.	Focuses mainly on energy management.	Integrate Digital Twin for comprehensive campus sustainability analytics.

From Building Information Modelling to Digital Twins: Digital Representation for a Circular Economy	BIM-to-Digital Twin transformation framework.	BIM lifecycle datasets.	Better lifecycle management and digital representation.	Limited discussion on cloud-based implementation.	Integrate IoT devices and real-time cloud synchronization.
Digital Twin-Enabled Smart Facility Management: A Bibliometric Review	Bibliometric analysis of Digital Twin research in facility management.	Published research articles.	Highlights research trends and future directions.	No real-world implementation.	Develop practical cloud-based Digital Twin applications for campuses.
A Review of Building Digital Twins to Improve Energy Efficiency	Systematic review of Digital Twin technologies for energy-efficient buildings.	Published research studies.	Comprehensive review of energy optimization approaches.	Focuses on buildings rather than campus-wide systems.	Extend Digital Twin to smart campus sustainability analytics.
AI-Driven Digital Twins for Enhancing Indoor Environmental Quality and Energy Efficiency in Smart Building Systems	AI-enabled Digital Twin for indoor environmental monitoring and energy optimization.	Indoor environmental and energy consumption datasets.	Improves energy efficiency, fault detection, and occupant comfort.	Limited to individual buildings.	Extend to multi-building smart campus Digital Twin platforms.